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Sample transfer device

Abstract

A device for collecting and transferring liquid samples having a detachable sample collecting portion includes an elongated body with a liquid collecting portion disposed at one end and a handling portion disposed at the other end separated by a frangible portion. The sample collecting portion includes an inner surface configured to be wetted by the sample. After collection, the sample is transferred into a separate housing by inserting the sample collecting portion filled with sample into a confining cavity and subsequently breaking the sample transfer device along the frangible portion. The liquid sample is extracted from the collecting portion using known means to drive and control flow in fluidic devices including rotation, capillary pressure or pneumatics. The separate housing may be part of a device capable to process the liquid sample to provide for a system for handling liquid samples.

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Background/Summary

RELATED APPLICATIONS

(1) The present application is a National Phase entry of PCT Application No. PCT/EP2019/060144, filed Apr. 18, 2019, which claims priority from PT Application No. 110702 K filed Apr. 18, 2018, both of these disclosures being hereby incorporated by reference in their entirety.

FIELD

(2) The present disclosure relates to a sample transfer device comprising a detachable sample collecting portion, sets of attached sample transfer devices and methods of manufacturing the same. The present disclosure further relates to methods of using the sample transfer device to collect and transfer samples to a separate housing, systems for handling liquids employing the sample transfer device in combination with devices for processing liquid samples and methods for processing liquid samples using such systems.

BACKGROUND

(3) Analytical devices are now widespread for a variety of applications in the laboratory, home and field use. Recently, advances in materials and manufacturing technologies prompted a generalized trend in device miniaturization and there are now a number of products in commercialization employing small foot-print analytical devices, many of which, consist of single use disposable cartridges. The latter may feature embedded reagent chemistries and detection capabilities or alternatively, designed to be integrated in a second device for assisting with diverse functions, such as sample processing or detection of specific substances.

(4) A key aspect of usability of many analytical devices and in particular miniaturised devices is the presentation of the sample to the device. Many devices require the sample to be a liquid or a liquid preparation of the original sample. For example, products used in diagnostics such as lateral flow devices common in pregnancy test kits or blood glucose strips used in the management of diabetes employ materials and constructions which wick small drops of the intended samples exploring what is known in the art as imbibition and wetting or capillary filling. Such effects are also used in other devices, in particular those employing designed micro-architectures in which the balance between two dominant forces acting on the liquid samples, such as gravity and surface tension, trade-off as the ratio between surface area and volume of the confined sample increases. These constitute an important and rapid growing class of devices known in the art as microfluidic devices, which display intricate arrangements of fluidic passages having at least one dimension in the sub-millimeter scale. For example, some of these devices are based upon recent concepts of lab on a chip, lab on a card, lab on a strip or lab on a disc, which integrate a variety of functions for processing and analyzing samples and are constructed in small dimensions, or at least, thin geometries.

(5) The widespread use of miniaturized, or otherwise thin, devices in a variety of settings present significant challenges in sample handling and management. First, the sample volumes are relatively small (typically microliters) and thus require specialized devices such as micro-pipettes or capillary collection tubes. These are common in laboratories but are less usual at home or field settings. Even with such devices provided by manufacturers, transferring small samples, through small apertures in miniaturized devices, is prone to operator error often resulting in compromised results or poor user experience.

(6) Inspired by the examples of lateral flow technology or blood glucose strips, some disposable cartridges are provided with wicking or capillary filling capabilities to allow for direct sample collection using the analytical device by touching a surface of the liquid sample. Despite being

simple in concept, this choice requires the device to contact potentially contaminated or hazardous samples and cleaning excess sample is not always possible as this can also remove or contaminate the intended liquid sample volume. As a consequence, additional measures or device complexity are required in order to contain the hazard of devices contaminated with samples at the external surface. In addition, this choice is also very limiting with respect to collecting samples from narrow containers such as vials (for example containing quality control solutions to ascertain analytical device condition) or test tubes from earlier collected samples, in which cases, a transfer device is still needed. Transfer devices used for these purposes are also potentially contaminated and need to be discarded according to the applicable Health and Safety Regulations.

SUMMARY

(7) Aspects of the disclosure are set out in the independent claims. Further, optional features of embodiments are set out in the dependent claims.

(8) In overview, liquid sample transfer devices featuring detachable sample collecting portions, sets of attached sample transfer devices and methods of manufacturing the same are disclosed. Also disclosed are methods of using sample transfer devices to collect and transfer samples to separate housings, systems for handling liquids employing the sample transfer devices and methods for processing liquid samples using such systems.

(9) In one aspect, a sample transfer device for receiving and holding a liquid sample is disclosed. The device comprises an elongate body having a first end opposed to a second end in the elongate direction, a handling portion disposed at the first end and arranged to provide for gripping and handling means and a sample collecting portion disposed at the second end and arranged to provide for means to receive a volume of a liquid sample. The device further comprises a frangible portion disposed between the handling portion and the sample collecting portion, wherein the frangible portion is configured to enable complete separation of the sample collecting portion from the sample transfer device.

(10) The elongate body allows for collecting samples with a comfortable distance between the handling portion and the liquid sample. This prevents contaminations, from or to, the liquid sample, and exposure of the user to potentially hazardous substances present in the sample. Samples to be handled with the sample transfer device may be of diverse types and origins such as, for example, biological samples, environmental samples, food samples, or cosmetic samples.

(11) The frangible portion typically comprises a reduction of the device resisting cross section configured to resist mechanical loading during sample collection and handling when the device is mostly unconstrained, such as for example bending of the elongate body. The main purpose of the resisting cross section reduction is to promote rupture or fracture when the device is constrained and mechanically loaded, as it favours localised deformation at the frangible portion. The reduction in device resisting cross section may be notch or a through cut to yield separate cross sections of smaller dimensions. The mechanical deformation along the frangible portion leading to separation of the sample collecting portion from the device may be achieved, for example, by bending, tearing or twisting the handling portion relative to the sample collecting portion. Preferably, the sample collecting portion is confined or supported on a separate surface or surfaces prior to separation. In other words, the sample collecting portion is secured against movement in at least one direction, when the handling portion is moved partially or totally in that same direction.

(12) The frangible portions according to the present disclosure may comprise one or more perforations through the thickness of the elongate body. In some embodiments, the perforation extends in at least two directions perpendicular to the thickness direction. In other embodiments, the frangible portion comprises at least two perforations through the thickness of the device. The perforations may have different shapes or the same shape disposed in different orientations. In some other cases, all perforations extend in at least two directions perpendicular to the thickness direction.

(13) The sample collecting portion according to the present disclosure, typically comprises a

surface configured to be wetted by the liquid sample. For example, the surface may be hydrophilic or lipophilic in nature, i.e., wetted by water as opposed to be wetted by oils and fats, or vice versa. In other embodiments, the surface is defined as the internal structure of porous media pads such as for example filter pads, membrane pads and woven or non-woven fabric pads. In these embodiments samples are absorbed by the porous material touching the liquid surface, and the overall pad dimensions and internal pore structure determine the maximum volume of sample collected.

(14) In some embodiments, the sample collecting portion comprises at least one cavity wherein the at least one cavity comprises at least one aperture serving as inlet port for the liquid sample and at least one other aperture to act as vent as the sample fills the cavity. In preferred embodiments, the cavity is defined between two opposite and substantially parallel inner surfaces, wherein one or both inner surfaces are configured to be wetted by the liquid sample. In this manner, the samples are drawn into the cavity of the sample collecting portion simply touching the liquid surface with the aperture serving as liquid inlet port. The cavity geometry may also be defined as a perforation through a spacer layer positioned between a first and second outer layers, each of which define an inner surface of the cavity. The maximum sample volume collected is determined by the cavity geometry and dimensions and the position of the apertures serving as inlet and vent. In some embodiments, the inlet port and air vent are substantially coincident with a perforation through the first or second outer layers. In some other embodiments, the perforation or perforations through the first and second outer layers are perfectly aligned.

(15) In some embodiments, the sample collecting portion may comprise more than one surface configured to be wetted by the liquid sample to collect more than one volume of a liquid sample or samples. In this manner, it is possible to use the same device to collect and transfer more than one volume of a liquid sample or samples. For example, the sample collecting portion may be arranged with two separate liquid absorbent pads or two separate cavities.

(16) In another aspect, sets of attached sample transfer devices are disclosed. The set comprises a plurality of sample transfer devices connected by frangible connections between adjacent pairs. The frangible connections are broken to separate individual sample transfer devices from the set, in order to be used. The frangible connections between sample transfer devices are preferably disposed in the handling portion as to avoid premature damage or weakening of the sample transfer device frangible portion.

(17) In another aspect, a method of manufacturing a plurality of sample transfer is provided. The method comprises receiving material as sheets, structuring the sheets to provide perforations or resizing width, aligning and bonding the different sheets, and finally cutting though the layered construction to yield either separate sample transfer devices or sets of attached sample transfer devices with frangible connections between adjacent pairs. The sheets of different materials may be provided as discrete units or in roll format.

(18) In another aspect, methods of using a sample transfer device for collecting and transferring liquid samples to a separate housing are disclosed. These involve contacting a liquid with the sample transfer device to collect the sample by wetting at least one surface of the sample collecting portion, inserting the sample collecting portion of the sample transfer device into the housing to confine the sample collecting portion against movement in at least one direction, breaking the sample transfer device along the frangible portion to leave the sample collecting portion filled with sample inside the housing, and subsequently driving flow of the liquid sample out of the sample collecting portion. The method may include an additional cleaning step after collecting the sample and before inserting the sample collecting portion of the sample transfer device into the confining housing. The flow of the liquid sample out of the sample collecting portion may be driven by a differential in pressure generated using a variety of means such as for example, gas pressure, capillary pressure or centrifugal pressure.

(19) In one other aspect, systems for handling liquids using the sample transfer device are

disclosed. The systems comprise the sample transfer device according to the present disclosure in addition to a device for processing liquid samples and means to drive and control liquid flow inside the device for processing liquid samples. The latter is provided with at least one aperture in communication with at least one receiving cavity, wherein the at least one receiving cavity is configured to receive and retain the sample collecting portion of the sample transfer device. The receiving cavity in the device for processing liquid samples is arranged to provide for a tight (or interference) fit in at least one direction to secure the sample collecting portion during separation of the handling portion. When the sample collecting portion of the sample transfer device is inserted and secured in position, the frangible portion may be preferentially positioned near or at an edge of the device for processing samples aperture, as to further facilitate rupture along the frangible portion.
(20) In some embodiments, it may be advantageous to arrange the receiving cavity of the device for processing liquid samples as an undercut extending laterally from the aperture. In other words, the cavity is disposed substantially parallel and near to an external surface of the device for processing liquid samples. This arrangement is particularly useful for devices for processing liquid samples configured in thin formats such as for example, lab on chip, lab on a disc or lab on card devices, wherein the device thickness is typically much smaller than other device dimensions, and the allowed penetration along thickness is minimal. In these cases, the sample transfer device must exhibit flexibility as it needs to bend in order to slide in the laterally disposed undercut and the frangible portion configured to resist the bending stresses without inducing any significant deformation leading to premature detachment of the sample collecting portion.
(21) For simplicity of use, it may be advantageous to provide an aperture in the device for processing liquid samples larger than the sample collecting portion cross section, and for the cavity to be arranged to continuously narrow with smooth surfaces to self-guide the sample collecting portion into a fit position. Alternatively, the sample collecting portion may be narrower at the tip which first enters the device for processing samples, and enlarge to a cross section tight fitting the cavity in at least one direction as it reached its intended position inside the device for processing samples.
(22) In another aspect, methods for collecting and processing a liquid sample using systems for handling liquid samples are disclosed. These involve collecting a liquid sample using the sample transfer device by wetting at least one inner surface of the sample collecting portion with the liquid sample, inserting the sample collecting portion of the sample transfer device into the device for processing liquid samples, breaking the sample transfer device along the frangible portion to leave the sample collecting portion filled with the liquid sample inside the device for processing liquid samples; and generating a differential pressure to drive flow of the liquid sample out of the sample collecting portion and into the device for processing liquid samples; and controlling the differential pressure to drive liquid flow inside the device for processing liquid samples to process the liquid sample.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) Specific embodiments are now described to illustrate aspects of the disclosure and by way of example with reference to the accompanying drawings, in which:
(2) FIG. **1** illustrates a sample transfer device;
(3) FIG. **2**A to FIG. **2**E illustrate use of the sample transfer device;
(4) FIG. **3**A to FIG. **3**G illustrate alternative configurations of the frangible portion in accordance with different embodiments of the sample transfer device;
(5) FIG. **4**A to FIG. **4**B illustrate a configuration of the sample collecting portion in accordance

with embodiments of the sample transfer device;
(6) FIG. **5**A to FIG. **5**B illustrate another configuration of the sample collecting portion in accordance with embodiments of the sample transfer device;
(7) FIG. **6** illustrates an alternative configuration of the sample collecting portion in accordance with embodiments of the sample transfer device;
(8) FIG. **7** illustrates another alternative configuration of the sample collecting portion in accordance with embodiments of the sample transfer device;
(9) FIG. **8**A to FIG. **8**F illustrate a specific configuration of the sample transfer device;
(10) FIG. **9** illustrates a set of attached sample transfer devices;
(11) FIG. **10** illustrates a method of manufacturing a plurality of sample transfer devices;
(12) FIG. **11** illustrates a method for collecting and transferring a liquid sample using the sample transfer device;
(13) FIG. **12** illustrates a system for handling liquids comprising the sample transfer device;
(14) FIG. **13**A to FIG. **13**C illustrate a specific configuration of a device for processing liquid samples comprised in the system for handling liquids;
(15) FIG. **14** illustrates a method for collecting, transferring and processing a liquid sample using the system for handling liquids;
(16) FIGS. **15**A to **15**E illustrate a further specific configuration of a sample transfer device; and
(17) FIG. **16** illustrates a method of manufacturing a sample transfer device.

DETAILED DESCRIPTION

(18) The disclosure provides a sample transfer device comprising a detachable sample collecting portion. Methods of using the sample transfer device to collect and transfer samples to a separate housing, systems for handling liquids employing the sample transfer device in combination with devices for processing liquid samples and methods for processing liquid samples using such systems, are also disclosed.

(19) With reference to FIG. **1**, a sample transfer device **100** for receiving and holding a liquid sample is described. The sample transfer device **100** comprises an elongate body having a first end **101** opposed to a second end **102** in the elongate direction **111**. A handling portion **103** disposed at the first end **101** is arranged to provide for gripping and handling means. A sample collecting portion **104** disposed at the second end **102** is arranged to provide for means to receive a volume of the liquid sample. A frangible portion **105** disposed between the handling portion **103** and the sample collecting portion **104** is configured to enable complete separation of the sample collecting portion **104** from the sample transfer device **100**. In preferred embodiments, the frangible portion **105** is configured to break when the sample collecting portion **104** is secured and the sample transfer device **100** is mechanically deformed along the frangible portion **105**.

(20) In use, the elongate body of the sample transfer device **100** allows for a comfortable distance between the handling portion **103** and the liquid sample during collection and transfer of the sample. This prevents contaminations, from or to the liquid sample, and exposure of the user to potentially hazardous substances present in the sample. In preferred embodiments, the distance of the frangible portion **105** to the first end **101** of the elongated body, i.e. the length of the handling portion **103**, is larger than the distance of the frangible portion **105** to the second end **102** of the elongate body, where the sample collecting portion **104** is positioned. In another preferred embodiment, the distance of the frangible portion **105** to the first end **101** is at least three times the distance of the frangible portion **105** to the second end **102**. It will be apparent to the person skilled in the art that it is possible to increase the length of the handling portion **103** using extensions with additional parts connecting to the handling portion **103** of the sample transfer device **100**. For these alternatives, the length of the handling portion is considered as the full length in the elongate direction **111** including the extension, or extensions, used.

(21) Samples to be handled with the sample transfer device **100** may be of diverse types and origins such as, for example, biological samples including urine, blood, serum, sputum and saliva,

environmental samples such as water and air condensates, food samples such as beverages and diverse sauces, or cosmetic samples such as fragrances and emulsions. Samples of solid nature can also be handled with the sample transfer device **100**, provided a first liquefaction step is performed in order to produce a liquid flowing medium such as for example a solution, a suspension or an emulsion.

(22) Operation of the sample transfer device **100** is now described with reference to FIG. **2**A to FIG. **2**E. In a first step, the sample collecting portion **104** of the sample transfer device **100** contacts a liquid surface. The liquid **206** can be presented as a drop (FIG. **2**A) such as for example, a blood drop formed from a finger prick or a condensation drop formed on a cold surface. Alternatively, the liquid **206** may be presented inside a container **207** (FIG. **2**B), such as for example a vial or a test tube. A liquid sample is drawn into the sample collecting portion **104** of the sample transfer device **100**.

(23) Once the liquid sample has been collected, it is retained inside the sample collecting portion **104** during handling. In a subsequent step, the sample transfer device **100** is mechanically deformed along the frangible portion **105** until rupture or fracture. At this point the sample collecting portion **104** is completely separated from the sample transfer device **100**. The mechanical deformation along the frangible portion **105** leading to breaking of the sample transfer device **100** and separation of the sample collecting portion **104** from the sample transfer device **100** may be achieved, for example, by tearing (illustrated in FIG. **2**C as movement **203**), folding back (illustrated in FIG. **2**D as movement **213**), or twisting (illustrated in FIG. **2**E as movement **223**) the handling portion **103** relative to the sample collecting portion **104**. Preferably, the sample collecting portion **104** is secured against movement in at least one direction, when the handling portion **103** is moved partially or totally in that same direction. This can be achieved, for example, inserting the sample collecting portion **104** into a slot or into a confining housing **250** prior to mechanically deforming the sample transfer device **100** along the frangible portion **105** to promote complete separation of the sample collecting portion **104** from the sample transfer device **100**.

(24) The frangible portion **105** typically comprises a reduction in the resisting cross section of the sample transfer device **100**. Additionally, the frangible portion **105** is configured to resist mechanical loading during sample collection and handling when the sample transfer device **100** is mostly unconstrained and also to resist mechanical stresses arising during insertion into a slot or confining housing **250**. The reduction in the resisting cross section of the sample transfer device **100** may be provided as a notch or as a through cut to yield separate cross sections of smaller dimensions. The main purpose of the reduction in resisting cross section is to favour localised deformation leading to rupture at the frangible portion **105**, when the sample collecting portion **104** is constrained against movement in at least one direction and the sample transfer device **100** is mechanically loaded in that direction.

(25) Alternative configurations of the frangible portion **105** in accordance with different embodiments of the sample transfer device **100** are now described with reference to FIGS. **3**A to **3**G. The frangible portion **105** may comprise one or more perforations through the thickness of the elongate body. Thickness is to be understood as the smallest dimension of the sample transfer device **100** which is typically perpendicular to the elongate direction **111**. Thickness direction **311**, is to be understood as the direction along the thickness. Moreover, perforations are to be understood as cuts through the thickness and may or may not involve removal of material. For example, perforations may be punctures or linear cuts without material removed or cut out areas of a given shape with material removed.

(26) In some embodiments (FIG. **3**A), the frangible portion **105** comprises one perforation **308** through the thickness of the elongate body extending in at least two different directions perpendicular to the thickness direction **311**. In other embodiments, the frangible portion **105** comprises at least two perforations through the thickness of the elongate body of the sample transfer device **100**. FIG. **3**B illustrates one such embodiment in which perforation **309** extends in a

first direction perpendicular to the thickness direction **311** (out of the plane), and perforation **310** extends in a second direction also perpendicular to the thickness direction **311** (out of the plane) and different from the first direction. FIG. **3**D illustrates other such embodiment in which each of the at least two perforations **311** and **312** extends in at least two directions perpendicular to the thickness direction **311** (out of the plane). In yet further embodiments (FIG. **3**D to FIG. **3**G), the frangible portion **105** comprises multiple perforations. In some embodiments, the perforations may have different shapes or the same shape disposed in different orientations.

(27) Alternative configurations of the sample collecting portion **104** in accordance with different embodiments of the sample transfer device **100** are now described with reference to FIG. **4** to FIG. **8**. The sample collecting portion **104** comprises at least one surface configured to be wetted by the liquid sample. In some embodiments, the surface configured to be wetted by the liquid sample may be hydrophilic, i.e., wetted by aqueous media. In other embodiments the surface may be lipophilic, i.e., wetted by oils and fats. In other embodiments, the surface configured to be wetted by the liquid sample is comprised within a liquid absorbent material. For example, the surface configured to be wetted by the liquid sample may be defined as the internal structure of porous material such as, for example, filter pads, membrane pads and woven or non-woven fabric pads. A pad is to be understood as a flat piece of a liquid absorbent material. In such embodiments, samples are absorbed by the porous material and the overall pad dimensions and internal pore structure determine the maximum volume of sample collected. In some embodiments, a cavity with a wall providing a hydrophilic surface may be combined with such a pad.

(28) FIG. **4**A illustrates a sample transfer device **400** comprising a handling portion **103**, a sample collecting portion **104** and a frangible portion **105**. FIG. **4**B illustrates a cross sectional view of the sample transfer device **400** along the elongate direction **411**. The sample collecting portion **104** comprises a surface **421** configured to be wetted by the liquid sample. Surface **421** is comprised within a liquid absorbent pad.

(29) In other embodiments, the sample collecting portion **104** of the sample transfer device **100** comprises at least one cavity. The at least one cavity comprises at least one aperture serving as inlet port for the liquid sample and at least one other aperture to act as an air vent as the sample fills the cavity. FIG. **5**A illustrates a sample transfer device **500** comprising a handling portion **103**, a sample collecting portion **104** and a frangible portion **105**. The sample collecting portion **104** comprises a cavity **522**. Cavity **222** comprises an inlet port **523** and an air vent **524**.

(30) In some of these embodiments, the at least one cavity is defined between two opposite and substantially parallel inner surfaces, wherein one or both inner surfaces are configured to be wetted by the liquid sample. FIG. **5**B illustrates a cross sectional view of the sample transfer device **500** along the elongate direction **511**. The sample collecting portion **104** comprises a cavity **522** defined between two opposite and substantially parallel inner surfaces **533** and **534**. Cavity **522** further comprises an inlet port **523** and an air vent **524**. Provided one or both inner surfaces **533** and **534** are configured to be wetted by the liquid sample, the liquid sample is drawn by capillary action through the inlet port **423** into the cavity **422** when the sample collecting portion **104** is brought in contact with a liquid surface.

(31) The geometry of cavity **522** may be defined as a perforation through spacer layer **530** positioned between a first **531** and second **532** outer layers. Each of the outer layers **531** and **532** define inner surfaces **533** and **534**, respectively, of the cavity **522**. The maximum sample volume collected is determined by the geometry and dimensions of the cavity **522** and the position of the apertures serving as the inlet port **523** and the air vent **524**. The different layers of the sample transfer device **500** may, in some embodiments, be arranged in a laminate comprising additional adhesive material (intermediate) layers **535** to bond the first **531** and second **532** outer layers to the spacer layer **530**.

(32) In yet further embodiments of the sample transfer device **100**, the sample collecting portion **104** may comprise more than one surface configured to be wetted by the liquid sample to allow for

collecting more than one volume of a liquid sample or different liquid samples. In this manner, it is possible to use the same sample transfer device to transfer more than one volume of a liquid sample or different samples.

(33) FIG. **6** illustrates a sample transfer device **600** comprising a handling portion **103**, a sample collecting portion **104** and a frangible portion **105**. The sample collecting portion **104** comprises two separate liquid absorbent pads **641** and **661**. FIG. **7** illustrates a sample transfer device **700** comprising a handling portion **103**, a sample collecting portion **104** and a frangible portion **105**. The sample collecting portion **104** comprises two separate cavities **742** and **762**. Each one of these cavities having a respective inlet port **743** and **763** and a respective air vent **744** and **764**. Optionally, the same air vent may be shared by the two cavities **742** and **762** (not shown).

(34) Other preferred embodiments of the sample transfer device will now be described with reference to FIGS. **8**A to **8**E. FIG. **8**A illustrates a sample transfer device **800** for receiving and holding a liquid sample. The sample transfer device **800** comprises an elongate body having a first end **101** opposed to a second end **102**. A handling portion **103** disposed at the first end **101** is arranged to provide for gripping and handling means. A sample collecting portion **104** disposed at the second end **102** is arranged to provide for means to receive a volume of the liquid sample. A frangible portion **105** disposed between the handling portion **103** and the sample collecting portion **104** is configured to enable complete separation of the sample collecting portion **104** from the sample transfer device **800**. The frangible portion **105** is configured to break when the sample collecting portion **104** is secured and the sample transfer device **800** is mechanically deformed along the frangible portion **105**.

(35) With reference to FIG. **8**B, the sample collecting portion **104** of the sample transfer device **800** comprises a cavity **822** of a substantially circular shape. This improved configuration facilitates capillary filling and at the same time provides a familiar shape to determine sample sufficiency. The cavity **822** is defined by two opposing and substantially parallel inner surfaces provided by a first outer layer **831** (FIG. **8**C) and a second outer layer **832** (FIG. **8**E) and a perforation through a spacer layer **830** (FIG. **8**D) disposed intermediate the outer layers. The thickness of the spacer layer **830** is preferably below 1 mm or even below 0.5 mm. It should be noted that confining a given volume in thinner cavities results in larger sample areas which are easier to analyse with regard to sample sufficiency. It may also be advantageous that the first outer layer **831** and the second outer layer **832** are provided in a transparent material. Alternatively, the respective one or both inner surfaces may exhibit a certain haze to transparency, and loose haziness as the cavity **822** is filled with liquid. This latter effect is particularly useful when handling transparent samples as it helps to ascertain the filling of cavity **822**.

(36) FIG. **8**F illustrates a cross sectional view of the sample transfer device **800** along the elongate direction **811**. The air vent **824** is substantially coincident with perforations **871** (FIG. **8**C) and **872** (FIG. **8**E) through the outer layers **831** and **832**, respectively. In improved configurations, there is a partial or total overlap between perforations **871** and **872**. Similarly, the inlet port **823** is substantially coincident with perforations **881** (FIG. **8**C) and **882** (FIG. **8**E) through the outer layers **831** and **832**, respectively. Also in this case, a partial or total overlap between the perforated areas of both layers is advantageous. The overlap between perforations **871** and **872** and perforations **881** and **882** prevents the liquid sample to be removed from the cavity **822** through the inlet port **823** or through the air vent **824**, when a tissue paper or other absorbing medium is used to clean the external surface of the sample transfer device **800**. In these cases, it may be further advantageous that both inner surfaces of the first and second outer layers **831** and **832** are wetted by the liquid sample.

(37) With reference to FIG. **9**, a set of attached sample transfer devices **900** comprises a plurality of sample transfer devices **800** arranged in a row. It will be appreciated that sample transfer devices **100**, **400**, **500**, **600** and **700** can be arranged in a similar fashion. In set **900**, pairs of adjacent sample transfer devices **800** are connected to each other by at least one frangible connection **910**.

The frangible connection **910** may be broken to separate a sample transfer device **800** from the set **900** to enable it to be used. The frangible connection **910** may preferably be positioned in the handling portion **103** of the sample transfer device **800** away from the frangible portion **105** to avoid mechanical deformation or premature weakening of the frangible portion **105**.

(38) In some embodiments, the plurality of sample transfer devices in set **900** do not need to be identical. The set **900** may comprise a plurality of sample transfer devices arranged in a plurality of rows and columns. In any given row, pairs of adjacent sample transfer devices are connected to each other by at least one frangible connection **910**. Between rows pairs of adjacent sample transfer devices are connected to each other by at least one frangible connection **910**.

(39) A method of manufacturing a plurality of sample transfer devices **500**, **700** and **800** is now described with reference to FIG. **10**. The method **1000** comprises in a first step **1002** receiving one sheet of spacer material, two sheets of outer layer material, and two sheets of adhesive material to be disposed intermediate the outer layer material and the spacer material in the final laminate. One or both of the outer layer material sheets have at least one side configured to be wetted by the liquid sample. For example, one side having been pre-treated to exhibit a hydrophilic character. At step **1004**, the adhesive sheets are first applied on either side of the spacer material to form what is commonly referred to as a spacer tape. The spacer tape is then perforated to define the geometry of the cavity (or cavities) **522**, **742**, **762**, **822** of the sample collecting portion **104**. At this step, at least one sheet of the outer layer material is perforated to provide for the air vent (or vents) **524**, **744**, **764**, **824** of the sample collecting portion **104**. Optionally, further perforations on the same or both sheets of outer layer material may be performed in the same cutting action. For example, additional perforations for inlet ports **523**, **743**, **763**, **823** and frangible portion **105** may also be executed at this step.

(40) Subsequently, at step **1006**, the sheets of outer layer material with the hydrophilic side oriented towards the spacer tape are aligned and applied to the spacer tape to form a laminate construction comprising a plurality of uncut sample transfer devices. The plurality of uncut sample transfer devices may be arranged in various ways, for example in one or more rows or one or more orientations. A final cutting step **1008** through the laminate thickness, then yields either separate sample transfer devices or sets of attached sample transfer devices with frangible connections between adjacent pairs as described with reference to FIG. **9**. It is also possible to perform the perforations of the frangible portion **105** of the sample transfer devices only in step **1008**, in the same cutting action. In some embodiments, the spacer tape is provided with a perforation for the cavity (or cavities), and laminated together with the outer layers. Subsequent to this lamination step, perforations are then made in the resulting laminate to form the air vent and inlet port by perforating through all five sandwiched layers (outer—intermediate—spacer—intermediate—outer), thereby ensuring alignment of the perforations for the inlet port and air vent.

(41) Method **1000** is equally applicable for manufacturing sample transfer devices **400** and **600**, comprising a liquid absorbent material **421**, **641**, **661** in the sample collecting portion **104**. In step **1002** the method comprises receiving one sheet of a backing material (for example the same as the spacer material), one sheet of adhesive material and one sheet of liquid absorbent material. In step **1004**, the two sheets of adhesive and liquid absorbent material **510** are cut through thickness to provide strips having substantially the same width. The two strips are aligned together and applied to the backing material sheet in step **1006**. Similar to the above description, a final cutting step **1008** then yields either separate sample transfer devices or sets of attached sample transfer devices with frangible connections between adjacent pairs. At this step the perforations of the frangible portion **105** are also executed in the same cutting action.

(42) The application of the different material layers to each other may be carried out by roll lamination or flat press bonding either with discrete sheets or with roll to roll converting systems. Perforations as well as final cuts may be provided by die cutting, kiss cutting or laser cutting either in batch or continuous processes. Preferably, all steps are integrated in a roll to roll process for

increased manufacturing efficiency, in which case the sheets corresponding to the different material layers are supplied in roll format. The manufacturing method **1000** enables quick and easy manufacture of a plurality of sample transfer devices. In those embodiments where frangible connections remain, the method provides a convenient way of supplying a plurality of attached sample transfer devices in multiple unit sets.

(43) With reference to FIG. **11**, a method **1100** of using the sample transfer device **100** for collecting and transferring liquid samples to a separate housing is now described. At step **1104**, a housing is provided to confine the sample collecting portion **104** of the sample transfer device **100**. Subsequently at step **1106**, the method **1100** further comprises contacting a liquid with the sample transfer device **100** to collect the sample by wetting at least one surface of the sample collecting portion **104**. At step **1108**, the sample collecting portion **104** of the sample transfer device **100**, is inserted into the housing to confine the sample collecting portion **104** and constrain movement in at least one direction. At step **1110**, a movement of the handling portion **103** of the sample transfer device **100** in the at least one direction induces mechanical deformation and breaks the sample transfer device **100** along the frangible portion **105** to leave the sample collecting portion **104** filled with sample inside the housing. The method **1100** may be performed using any of the sample transfer devices **400**, **500**, **600**, **700** and **800**.

(44) The method **1100** may include an additional step **1107**, for cleaning excess liquid sample from the outer surface of the sample transfer device **100** after collecting the liquid sample and before inserting the sample collecting portion **104** of the sample transfer device **100** into housing. This step is particularly useful when collecting samples from vials and test tubes as the sample collecting portion **104** may be dipped below the liquid surface and an excess of liquid wets the outer surface of the sample transfer device **100**, which may contaminate the housing during insertion.

(45) With reference to FIG. **12**, a system **1200** for handling liquid samples is now described. The system **1200** comprising the sample transfer device **100** and a device for processing liquid samples **1202**, a processor **1206**, a controller **1208** and means **1210** to drive liquid flow inside the device for processing liquid samples **1202**. Means to drive and control liquid flow inside devices for processing liquid samples are common in the art and include the use pneumatic or vacuum pumps to generate gas pressure differentials, DC motors to generate centrifugal pressure differentials by rotation, and surface tension gradients to generate capillary pressure differentials. The system **1200** may comprise any of the sample transfer devices **400**, **500**, **600**, **700** and **800**.

(46) The device for processing liquid samples **1202** is provided with an aperture **1204** in communication with at least one receiving cavity **1203**, wherein the at least one receiving cavity is configured to receive and retain the sample collecting portion **104** of the sample transfer device **100**. In some embodiments, the sample collecting portion **104** of the sample transfer device **100** and the receiving cavity **1203** of the device for processing liquid samples **1202** are configured to provide for a tight (or interference) fit in at least one direction in order to secure the sample collecting portion **104** during separation from the sample transfer device **100**. Preferably, when the sample collecting portion of the sample transfer device is inserted in the receiving cavity **1203** and secured in position, the frangible portion **105** is positioned near or at an edge of aperture **1204**, so as to further facilitate breaking the sample transfer device **100** along the frangible portion **105**.

(47) In other embodiments as illustrated in FIG. **13**A, the receiving cavity **1303** of the device for processing liquid samples **1302** is arranged as an undercut extending laterally from aperture **1304**. This ensures enough travel of the sample collecting portion **104** of the sample transfer device **100** into the device for processing liquid samples **1302** without requiring a significant penetration depth. In other words, the receiving cavity **1303** is disposed substantially parallel and near to an external surface of the device for processing liquid samples **1302**. Such embodiments are particularly useful for devices for processing liquid samples configured in thin formats such as for example, lab on chip, lab on a disc or lab on card devices, wherein the device thickness is typically

much smaller than other device dimensions. As a consequence, the allowed penetration depth is minimal and typically below 1 or 2 mm. In these cases, the sample transfer device **100** must exhibit flexibility as it needs to bend in order to slide in the laterally disposed undercut defining the receiving cavity **1303**. Additionally, the frangible portion **105** of the sample transfer device **100** is configured to resist the bending stresses to avoid premature detachment of the sample collecting portion **104** during insertion. Frangible portions **105** with perforations configured as shown in FIGS. **3**C to **3**G are fit for this purpose. The exact dimensioning of perforations and their spacing takes into account the mechanical properties of the sample transfer device constituent materials.

(48) In preferred embodiments as illustrated in FIG. **13**B, aperture **1304** in the device for processing liquid samples **1302** is larger than the sample collecting portion cross section to facilitate introduction of the sample collecting portion **104** into the receiving cavity **1303**. Additionally, the receiving cavity **1303** is arranged to continuously narrow with smooth surfaces **1305***a* and **1305***b* to self-guide the sample collecting portion into a fit position. Alternatively, the sample collecting portion **104** may be narrower at the tip which first enters the device for processing liquid samples **1302**, and enlarge to a cross section tight fitting the receiving cavity **1303** in at least one direction. An example of such a sample transfer device is illustrated in FIGS. **8**A to **8**F. One skilled in the art would combine some of the characteristics above to achieve the same end result of self-guiding the sample collecting portion into a secure position.

(49) FIG. **13**C illustrates an example of a device for processing liquid samples **1302** having a disc shaped body showing an aperture **1304** and a receiving cavity **1303** underneath the surface of the device. The device **1302** is arranged to rotate about the axis of rotation **1306** (out of the plane). In use, after sample collection, the sample collecting portion **104** of the sample transfer device **100** is inserted via aperture **1304** into the receiving cavity **1303**. The receiving cavity **1303** is configured for a tight fit with the sample collecting portion **104**. When the sample collecting portion is fit into position the frangible portion **105** of the sample transfer device **100** is substantially aligned with the edge of aperture **1304**. The handling portion **103** of the sample transfer device **100** is folded back along the frangible connection **105** to break the sample transfer device **100**, leaving the sample collecting portion **104** inside the receiving cavity **1303**. In preferred embodiments the receiving cavity **1303** is substantially aligned with the radial direction. Rotation of the device **1302** about the axis of rotation **1306** generates a differential pressure in the liquid sample to drive flow of the liquid sample out of the sample collecting portion **104** and into downstream structures of device **1302**, such as cavity **1307**.

(50) It will be apparent to a person skilled in the art that the receiving cavity **1303** may be positioned in a different orientation relative the radial direction. In other embodiments, the device for processing liquid samples **1302** could be arranged in a configuration other than a disk and the differential pressure to drive flow of the liquid sample out of the sample collecting portion **104** be achieved by other means. For example, a vacuum pump could be used to generate a vacuum pressure through cavity **1307** to extract the liquid sample. In yet another example, a liquid absorbing medium could be disposed in cavity **1307** to contact the inlet port of the sample collecting portion **104** in order to extract the sample by capillary pressure driven flow.

(51) With reference to FIG. **14**, a method **1400** for collecting, transferring and processing liquid samples is now described. At step **1402**, a system for handling liquids **1200** is provided. At step **1406**, the method **1400** further comprises contacting a liquid with the sample transfer device **100** to collect the sample by wetting at least one surface of the sample collecting portion **104**. At step **1408** the sample collecting portion **104** of the sample transfer device **100**, is inserted into the device for processing liquid samples **1202** to confine the sample collecting portion **104** and constrain movement in at least one direction. At step **1410**, a movement of the handling portion **103** of the sample transfer device **100** in the at least one direction induces mechanical deformation and breaks the sample transfer device **100** along the frangible portion **105** leaving the sample collecting portion **104** filled with sample inside the device for processing liquid samples **1202**. Step **1412**,

comprises generating a differential pressure to drive flow of the liquid sample out of the sample collecting portion **104** and into the device for processing liquid samples **1202**. The method **1400** further comprises step **1414** to control a differential pressure to drive liquid flow inside the device **1202** to process the liquid sample. The differential pressures to drive and control liquid flow in steps **1412** and **1414** may be generated by a variety of means common in the art, including the use pneumatic or vacuum pumps to generate gas pressure differentials, DC motors to generate centrifugal pressure differentials by rotation, and surface tension gradients to generate capillary pressure differentials. Steps **1412** and **1414** may employ different means to generate pressure differentials to drive and control liquid flow during the respective steps.

(52) The method **1400** may include an additional step for cleaning excess liquid sample from the outer surface of the sample transfer device **100** after collecting the liquid sample at step **1406** and before inserting the sample collecting portion **104** of the sample transfer device **100** into device **1202** at step **1408**. This additional step is particularly useful when collecting samples from vials and test tubes as the sample collecting portion **104** may be dipped below the liquid surface and an excess of liquid wets the outer surface of the sample transfer device, which may contaminate device **1202** during insertion. The method **1400** may be performed using any of the sample transfer devices **400**, **500**, **600**, **700** and **800**.

(53) Further embodiments of the sample transfer device will now be described with reference to FIGS. **15**A to **15**E. FIG. **15**A illustrates a sample transfer device **1500** for receiving and holding a liquid sample. The sample transfer device **1500** comprises a handling portion **103** disposed at the first end **101** and a sample collecting portion **104** disposed at the second end **102**. A frangible portion **105** disposed between the handling portion **103** and the sample collecting portion **104** is configured to enable complete separation of the sample collecting portion **104** from the sample transfer device **1500**.

(54) With reference to FIG. **15**B, the sample collecting portion **104** of the sample transfer device **1500** comprises a cavity **1522** of a substantially circular shape. The cavity **1522** is defined by two opposing and substantially parallel inner surfaces provided by a first outer layer **1531** (FIG. **15**C) and a second outer layer **1532** (FIG. **15**E) and a perforation **1570** through an inner spacer layer **1530** (FIG. **15**D).

(55) A perforation **1580** through the device thickness (FIGS. **15**C-**15**E) provides for the aperture **1523** (FIG. **15**B) to act as the inlet port of the cavity **1522** of the sample collecting portion. A perforation **1590** through the device thickness (FIGS. **15**C-**15**E) provides for the reduction of the device resisting cross section at the frangible portion **105**, thereby providing a weakened portion for controlled separation of the collecting and handling portions, and also provides for the aperture **1524** (FIG. **15**B) to act as the air vent of the cavity **1522** of the sample collecting portion. By providing the air vent opening in the frangible portion, the device can be manufactured in less steps or with less cuts and a more compact arrangement of the sample collecting portion is enabled.

(56) The use of single cuts through the device thickness to provide for both the inlet port and the air vent also prevents the liquid sample to be removed from the cavity **1522** through the inlet port **1523** or through the air vent **1524**, when a tissue paper or other absorbing medium is used to clean the external surface of the sample transfer device **1500**.

(57) With reference to FIG. **16**, a method of manufacturing a sample transfer device, for example as described above, comprises providing **1602** a spacer layer (comprising a single layer of material, laminate comprising a spacer material with an intermediate layer laminated to it one or both sides, or a laminate of any suitable number of layers), for example inner spacer layer **1530**, and perforating **1604** a cut-out through the spacer layer to form a cavity, for example perforation **1570**. The perforated spacer layer is then laminated **1606** on either side with a coven outer layer, for example outer layers **1531** and **1532**, in some embodiments with a hydrophilic surface of the outer layers facing the spacer layer. The resulting laminate is then perforated **1608** to form a first hole intersecting the cavity to provide the reduced cross-section of the frangible portion and the air vent,

for example perforation **1590**. The laminate is further perforated **1610** to form a second hole intersecting the cavity and disposed in a region of an inlet end of the sample transfer device to provide the inlet opening and perforated **1612** to cut out the outline of the sample transfer device (individually or as part of a connected set of sample transfer devices), with the intersection between the perforations and steps **1610** and **1612** forming a notch at the inlet end. Where desired, the outer layers may be cut **1614** through to the spacer layer, cutting any intermediate layers down to the spacer layer material if needed, in the frangible portion across the first hole. This may be useful to facilitate a clean break in the frangible portion, for example when the outer layer materials are malleable rather than brittle. It will be appreciated that steps **1608** to **1614** can be performed in any order and can be grouped to be performed simultaneously as convenient based on the design of the relevant cutting tools.
(58) It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other implementations will be apparent to those of skill in the art upon reading and understanding the above description. Although the present disclosure has been described with reference to specific example implementations, it will be recognized that the disclosure is not limited to the implementations described but can be practiced with modifications and alterations within the spirit and scope of the appended claims. Accordingly, the specification and drawings are to be regarded in an illustrative sense rather than a restrictive sense. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.
(59) For one example, embodiments of the sample transfer device comprising multiples cavities for collection of individual sample volumes can be configured in many different ways. Additionally, embodiments providing multiple apertures (i.e. multiple inlet ports) per cavity could be used, for example, to divide the volume of the liquid sample in multiple fractions when transferring the sample from the sample transfer device to a second device. These are considered within the spirit and scope of the present disclosure.

Claims

1. A sample transfer device comprising a sample collecting portion and a handling portion separated by a frangible portion, wherein the sample collecting portion comprises a cavity having a first opening at an inlet end of the sample transfer device to allow a sample liquid to be drawn into the cavity and a second opening to allow air to escape from the cavity as liquid is drawn into the cavity and wherein the frangible portion comprises a weak section of material with reduced resistance to breaking compared to adjacent sections of the sample transfer device adjacent to the weak section to facilitate separation of the sample collecting portion and the handling portion.
2. The sample transfer device according to claim 1, wherein the cavity has at least one surface presenting a hydrophilic material to the inside of the cavity.
3. The sample transfer device according to claim 1, wherein the cavity is defined by a through-hole in a spacer layer sandwiched between a respective outer layer on each side of the spacer layer, each outer layer forming a wall of the cavity.
4. The sample transfer device according to claim 3, wherein the spacer layer comprises a material exhibiting brittle fracture upon folding or shearing.
5. The sample transfer device according to claim 3, wherein the spacer layer comprises a material selected from the group of: Poly(methyl methacrylate), Polystyrene, Polylactic Acid, Polyvinyl Chloride, Polycarbonate, Styrene-acrylonitrile, Cyclic Olefin Copolymer.
6. The sample transfer device according to claim 3, wherein the spacer layer comprises Poly(methyl methacrylate).
7. The sample transfer device according to claim 3, wherein the spacer layer comprises a material capable of being provided as a roll for roll to roll processing.

8. The sample transfer device according to claim 3, wherein the first opening is provided by a notch in the inlet end of the sample transfer device extending through the spacer and outer layers.
9. The sample transfer device according to claim 3, wherein the second opening is provided by a hole extending through the spacer layer and outer layers.
10. The sample transfer device according to claim 9, wherein the hole is in the frangible portion to provide the weak section of material.
11. The sample transfer device according to claim 8, wherein the notch and/or hole have a rounded contour.
12. The sample transfer device according to claim 9, wherein the hole is substantially circular.
13. The sample transfer device according to claim 3, wherein the outer layers are cut across the frangible portion through to the spacer layer to provide the weak section of material.
14. The sample transfer device according to claim 3, wherein the at least one of the outer layers comprises a hydrophilic side facing the spacer layer.
15. The sample transfer device according to claim 14, wherein the spacer layer is sandwiched between an intermediate layer on each side and the laminate of the spacer and intermediate layers is sandwiched on each side between the outer layers, the cavity being formed by a through-hole through the intermediate and spacer layers.
16. The sample transfer device according to claim 1, wherein the sample transfer device is translucent or transparent in a first region coinciding with the cavity and opaque in a second region surrounding the first region.
17. The sample transfer device according to claim 16, wherein the sample transfer device is white in the second region.
18. The sample transfer device according to claim 15, wherein one or both of the intermediate layers are opaque.
19. The sample transfer device according to claim 18, wherein one or both of the intermediate layers are white.
20. The sample transfer device as claimed in claim 1, wherein an absorbent material is disposed inside the cavity.
21. The sample transfer device as claimed in claim 1, wherein the cavity has a substantially circular shape.
22. A kit comprising the sample transfer device as claimed in claim 1 and a microfluidic liquid sample processing device, wherein the microfluidic liquid sample processing device comprises an opening in an outer surface of the processing device giving access to the sample collecting portion to a space inside the processing device, wherein the opening comprises an edge and the space is dimensioned to accommodate the sample collecting portion with the frangible portion resting against or adjacent the edge when the sample collecting portion is fully inserted into the space.
23. A kit according to claim 22, wherein the liquid sample processing device comprises a centrifugal microfluidic liquid sample processing device having a body to be rotated about an axis of rotation to drive liquid flows inside the device, the device comprising an inlet port in a planar surface of the body intersecting the axis of rotation for receiving a sample transfer device, wherein the inlet port comprises an opening in the planar surface of the device giving access to the sample transfer device to an undercut below the planar surface defining an internal cavity to accommodate and hold in place a portion of the sample transfer device in an orientation substantially parallel to the planar surface and wherein the opening defines an edge for engagement with a portion of the sample transfer device.
24. A method of manufacturing a device as claimed in claim 3, the method comprising cutting through the spacer layer to form the cavity, making a first cut through the spacer layer and outer layers to form the first opening and making a second cut through the spacer layer and outer layers to form the second opening and frangible portion.